NAVIGATION / KEEP BESIDE THE MODULE

Quick reference card

From **Pads**, select a pad. Press **INFO** for its waveform and **INFO** again for pad parameters. From **Menu**, open **Tools** for global controls.

Second custom release / +c2

Pad tone	**Pad parameters > Tone.** Fidelity, Drive, Filter 2, Res 2 and Slope 1. Select Drive, Filter 2 or Res 2 and press INFO for modulation.
Clip timing	**Clip parameters > Pos.** Scroll to Loop Fade ms, Shift ms and Play Len. Sync now includes 1/32.
Beat repeats	**Clip > Pos > Live Flip.** For Play Len and Live Flip, enable Quant Size and choose a timed Sync division. Off rejoins the current beat.
Reverb	**FX > Reverb > Type.** Plate, Room, Hall or Chamber. Tap the right half of the grid to edit the extra controls.
Compressor	**Tools > Comp** for threshold, ratio, gain and parallel mix. **Tools > Mstr > Section > Comp env** for attack, release and on/off.
Master effects	**Tools > Mstr > Section > Master FX.** Distort, Drive dB and Sweep on Out 1/2. Sweep left for low-pass, right for high-pass.
Recording sends	**Selected pad > FX1 / FX2.** Set sends before recording. The live monitor uses them and the recorded input stays dry.
MIDI input log	**Tools > MIDI > Monitor.** Encoder 1 filters by channel. Freeze, then use encoder 2 to scroll. Back closes the log.

First custom release / +c

Waveform zoom	**Waveform.** Two fingers vertically change amplitude. Two-finger tap restores 100% vertical and fits the time axis.
Sample Speed	**Sample parameters > LFO/Spd > SPEED.** Select SPEED and press INFO for LFO/CV. Range 50-200%.
Display / reset	**Tools > Disp** for Scrn saver and Sig wake. Double-tap a numeric dial to restore its base value and keep its modulation.

FIRST CUSTOM RELEASE / +C

Waveform and display

These controls are included when you patch official 2.3.4 directly to +c2.

Set a loop boundary by eye

On a loaded pad, press INFO to see the waveform. Spread two fingers vertically to magnify amplitude, or pinch to reduce it. The range is **25-3200%**. Enlarge a quiet recording to find a crossing or reduce a loud one to bring its peaks back into view.

Horizontal finger spacing controls time zoom. A diagonal gesture can change both axes. A stationary **two-finger tap** returns amplitude to 100% and fits the waveform horizontally. Zoom changes the display only and lasts between gestures on that waveform.

Reset a dial

Double-tap a numeric dial on a pad or the Delay/Reverb grid to restore its base value. For example, Pitch returns to 0 semitones and Speed to 100%. Assigned modulation continues to move the effective value.

Blank the screen during long sessions

Under **Tools > Disp > Scrn saver**, select disabled or **1, 2, 3, 5, 10, 15, 30 or 60 minutes**. Disabled is the default. After the chosen idle interval the screen goes black while playback continues. The first touch, HOME/INFO press or encoder movement wakes it and is consumed. Release the control before editing.

Sig wake	Choose **disabled, MIDI, CV trig or both**. A qualifying signal wakes the screen and refreshes the idle timer. The first signal also performs its musical action.
MIDI / CV	MIDI Clock and Active Sensing are excluded from wake activity. CV wake uses rising trigger/gate events. Continuous changes, held gates and the separate clock input do not wake the screen.

The screen saver and wake preferences are saved globally. The original MIDI In tab is labeled **MIDI**.

FIRST CUSTOM RELEASE / +C

Speed and Pitch

Sample parameters > LFO/Spd > SPEED

Adjust playback speed without changing pitch

Drag SPEED or select it and turn its encoder. **50%** plays at half speed, **100%** uses native playback and **200%** plays at double speed. Live edits are supported. Select SPEED and press INFO to assign up to three native modulation sources and amounts, including LFO and external CV. Modulated Speed stays within 50-200%.

Which voice gets the stretch slot?

No active owner	One Sample voice can acquire Speed, or one or more voices can acquire native Pitch. If a new Sample voice requests both, Pitch takes priority.
Pitch active	Multiple native Pitch voices can play together. Active Pitch blocks Speed across the module.
Speed active	One stretched voice owns the slot. Pitch is blocked everywhere, including on that voice. Other voices can play at neutral speed.

You can prepare settings on any pad while the slot is busy. A voice started at neutral can acquire an available operation live. A voice refused a requested operation stays neutral for this playback instance. Restart it to try again. A modulation route can hold a reservation as it passes through neutral.

SPDBUSY / PITBUSY indicate that the requested operation is occupied. **RESTART** means this playback instance was refused. Your dial settings and modulation assignments are retained.

Speed applies to eligible forward Sample playback. Reverse, loop crossfade, active loop-boundary modulation and unsupported source configurations can fall back to native playback. Speed and its routes save with the pad. Earlier presets default to 100%.

SECOND CUSTOM RELEASE / +C2

Tone for every pad mode

Sample, Clip, Slice and Granular pad parameters > Tone

Choose the source character

Fidelity expands Interp. Normal and HighQ preserve the native choices. **SP1200, SP-12, S950, MPC60 and SP Raw** use sampler-inspired output coloration. Seven fixed-rate choices combine 12-bit quantization with **16, 12, 10, 8, 6, 4 or 2 kHz** rate reduction.

Drive and two filters in series

Drive	0-100% soft saturation before the pad filters. The default is 0%.
Slope 1	Select 12 or 24 dB/octave for the original Main-page Filter. The default is 12 dB/octave. Turn Filter away from center to hear the slope.
Filter 2 / Res 2	A second, independent 24 dB/octave filter. Negative values select low-pass, positive values select high-pass and center bypasses it. Res 2 controls its resonance and defaults to 50%.

Modulate Drive, Filter 2 or Res 2

Select the dial and press **INFO** to assign a source and amount. Each dial shows up to three source slots, subject to the native total of 12 saved modulation routes per pad. Back returns to the same Tone dial. Double-tap resets its base value. Fidelity and Slope 1 are manual selectors.

> On Tone, Fidelity and Drive sit above Filter 2 and Res 2 in the left bank. Slope 1 is at the top of the right bank. Tap a bank to give its dials the encoders.

The configuration tab is shortened to **Cnf**. On a blank pad, INFO twice now opens Sample parameters with the correct LFO/Spd tab and a working Speed dial at 100%.

SECOND CUSTOM RELEASE / +C2

Clip timing and Live Flip

Clip pad > INFO > INFO > Pos. Use the selection encoder to reach the lower rows.

Loop Fade ms	0-50 ms, default off. Dips the loop edges toward zero to reduce clicks. Start around 2-5 ms. The fade is capped at a quarter of a short loop.
Shift ms	-100 to +100 ms, default 0. Positive values start later in the file. Negative values bring the tail to the start. Shift wraps at the source boundary.
Play Len	Full or 1/4-32 bars in quarter-bar steps. Caps playback at the available source and rounds to the nearest Sync division, with a minimum of one division.
Sync	Adds 1/32. For Play Len and Live Flip, enable Quant Size and choose a timed Sync division such as 1/16 or 1/32.

Five ways to recut a four-beat bar

Off	The original phrase follows beats 1, 2, 3, 4.
Beat Rpt	Plays beats 1, 2, 2, 2.
1/8 Rpt / 1/16 Rpt	Beats 1-3 play normally. Beat four repeats its first eighth twice or its first sixteenth four times.
Half Rpt	Plays beats 1, 2, 1, 2.
Rotate	Plays beats 2, 3, 4, 1.

Live Flip stays selected until changed. Turn it Off during playback to rejoin the original phrase on the current beat. It works on mono and stereo Clips and uses no stretch voice. Set Beat Count to match the source. Re-cuts are hard edits, so a jump through sustained material may click.

Slice Mono now honors the saved Poly Mode after switching a polyphonic Sample pad to Slice. With Trigger, Mono and Play Thru On, a new slice cuts the previous one even with Quant Size set to None.

SECOND CUSTOM RELEASE / +C2

Reverb and master effects

Shape a send return under FX, or process the stereo mix under Tools.

FX > Reverb

Plate is the original algorithm and the default. **Room, Hall and Chamber** add Size, Diffusion and Width. Tap the right half of the grid to edit those controls. All four types offer Level. Its 50% position is unity, with about -12 dB to +12 dB available. Decay, Predelay and Damping retain their native modulation.

Type and reverb settings save in the preset. Changing Type briefly fades and clears the previous wet tail while the dry signal continues.

Tools > Comp

Threshold / Ratio	Threshold -48 to 0 dB. Ratio 1:1 to 20:1. Defaults are -4 dB and 4:1.
Gain / Mix	Gain -12 to +24 dB, default -4 dB. Parallel Mix runs from 0% dry to 100% processed, with 100% as the default.
Attack / Release	Under **Tools > Mstr > Section > Comp env**, set Attack to 1-200 ms and Release to 10-2000 ms. Defaults are 10 ms and 250 ms. The On/Off control operates the same compressor as Tools > Main.

Tools > Mstr > Section > Master FX

Select **Soft, Hard, Fold or Tube** distortion, or Off. Drive adds **0 to +24 dB** before the shaper. Soft rounds peaks, Hard clips, Fold folds peaks back, and Tube gives asymmetric saturation. Sweep works independently. Turn left for low-pass, right for high-pass and center for dry.

On Out 1/2 the order is **master distortion, sweep, then the optional compressor**. The mix includes the FX returns. Master settings save globally.

SECOND CUSTOM RELEASE / +C2

Recording sends and MIDI

Monitor a take through effects or inspect the messages reaching the module.

Record dry while monitoring through FX

Set FX1 and FX2 on the selected pad before recording. Both send levels survive the pad reset for recording. The input monitor uses the currently selected pad's send levels and follows the native Monitor On, Auto or Off setting. Select a different pad to change the monitor routing.

The WAV records the input, including its input gain. Delay and reverb heard through the sends stay out of the recorded sample.

Tools > MIDI > Monitor

Tap **Monitor** beside the version number. Capture starts on entry. The display shows seven rows, newest first, from a 32-event history. Read each row as elapsed time, input group, channel and message. For example, **0.350 A 03 CC 74=96** means group A received CC 74, value 96, on channel 3 after 350 ms.

Freeze / Resume	Freeze holds the visible history while capture and playback continue. Encoder 2 scrolls the frozen rows. Resume jumps to the latest events.
Ch All / Ch 1-16	Touch the channel button or use encoder 1 to filter the list. Transport messages appear on every channel setting.
Clear / Back	Clear restarts elapsed time and clears the log. Back stops capture and returns to the four MIDI settings.
CLK / SENSE	Separate lights indicate incoming MIDI Clock and Active Sensing. The rows show notes, CC, program changes, pressure, pitch bend and transport or reset messages.

A and B are native input groups. B includes the USB queue and one parser path, so the group does not uniquely identify a jack. SysEx and system-common messages are discarded before the monitor. Signal Wake is configured separately under Disp.

FIRMWARE UPDATE / BITBOX MKII ONLY

Install and return to stock

Browser patcher input: official 1010music Bitbox MkII firmware 2.3.4.

Make and install the file

1. Back up the entire microSD card, including projects, samples and presets. Download official 2.3.4 and extract the ZIP. Open the browser patcher and select the original **BITBOXMK2.bin**, including when upgrading from +c. It checks the input SHA-256 and verifies the output before download.

2. Copy the patched file to the card root. Its name must be exactly **BITBOXMK2.bin**. Remove any duplicate-download suffix such as (1). Power off, insert the card, and hold **INFO** while powering on. Release INFO when the upgrade message appears.

3. Allow about 30 seconds, keep power on and wait for automatic reboot. Press HOME until Menu appears and check the lower-right version, or read the top-right of Tools. Both should show **2.3.4+c2**.

Return to official 2.3.4

Replace the patched file in the card root with the original unmodified BITBOXMK2.bin from the official ZIP. Repeat the same INFO-held power-on procedure and wait for reboot. Confirm **2.3.4** in Menu.

SHA-256 fingerprints

Official 2.3.4 / 674,896 bytes

```
35ab7995ad0eca8484f3b28bc76389f10b2037dbb21a4a394e950d9329073e86
```

Custom 2.3.4+c2 / 727,440 bytes

```
8bcb28eb58be54147458dcf247b8f35bb3f15932bdc6bcf027c8da128283f895
```

Manual and video

Official MkII manual / Watch the first release (+c) overview. A video covering the second release (+c2) is coming soon.

Independent customization by vwal, offered as is. You install and use it at your own risk. To the extent permitted by applicable law, the author accepts no liability for hardware damage or for loss or corruption of projects, samples, presets or other data arising from installation or use.